

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Paper No. BPT-501: Mathematical Physics

Time: 3 Hours

Maximum Marks: 100

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **five** of the following:

[5 × 4 = 20]

(a) Evaluate ∇r^2 in spherical polar coordinates.(b) If A_{ij} is an antisymmetric tensor in a 3D space, then show that:

$$(\delta_{ik}a_j + a_ia_k)A_{ik} = 0$$

(c) Determine whether the following sets of functions are linearly independent:

(i) $\sin x, \sin 2x, \sin 3x$ (ii) $\sin x, \cos x, 1$ (d) If $xL_n''(x) + (1-x)L_n'(x) + nL_n(x) = 0$, find the second-order ordinary differential equation (2-ODE) satisfied by the function $y(x) = e^{-x/2}L_n(x)$. Also write down the orthogonality relation for Laguerre polynomials $L_n(x)$.(e) Given the matrix $A = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 4 & 2 \end{pmatrix}$, find the trace and determinant of A^{-1} without actually computing the inverse matrix A^{-1} . Outline your method.(f) Show that $\delta(kx) = \frac{1}{|k|}\delta(x)$, where $\delta(x)$ is the Dirac delta function.(g) The Hermite polynomials $H_n(x)$ can be obtained from the generating function $g(x, t) = e^{2xt-t^2} = \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n$. Show that:

$$\int_{-\infty}^{\infty} e^{-x^2} [H_n(x)]^2 dx = 2^n n! \sqrt{\pi}$$

Question 2.(a) Show that $g(x, t) = (1 - 2xt + t^2)^{-1/2}$ is the generating function for Legendre polynomials $P_n(x)$. [6]

(b) Consider the Bessel differential equation:

$$x^2 y'' + xy' + (x^2 - n^2)y = 0$$

(i) Identify its singularities. Determine whether they are regular or irregular singularities. [3]

- (ii) Solve this differential equation by the Frobenius method for $n = 1$ and $n = 0$. [8]
- (iii) Explain if a second linearly independent solution can be constructed using this method for $n = 1$. [3]

Question 3.

- (a) Consider a sphere of radius R with its center at the origin $(0, 0, 0)$. Draw neatly and label the spherical polar unit vectors $\hat{r}$, $\hat{\theta}$, $\hat{\phi}$ at the points $(R, 0, 0)$, $(0, R, 0)$ and $(0, 0, R)$, which are the points of intersection of the sphere with the right-handed Cartesian x, y, z axes. [8]
- (b) In spherical polar coordinates, the gradient of a scalar function $\psi(r, \theta, \phi)$ is given by:

$$\nabla\psi = \hat{r}\frac{\partial\psi}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial\psi}{\partial\theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial\psi}{\partial\phi}$$

Show that the Laplacian is:

$$\nabla^2\psi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2}$$

[12]

Question 4.

- (a) Explain the terms *polar vector* and *axial vector* with one physical example of each. [4]
- (b) Express the components of $\vec{E} = \vec{A} \times \vec{B}$ in terms of the Levi-Civita symbol ϵ_{ijk} and the components of $\vec{A}$ and $\vec{B}$. Use this tensor notation to verify the vector identity:

$$(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C})$$

[8]

- (c) Show that the transpose of a square matrix A (denoted A^T) has the same eigenvalues as the matrix A . [8]

OR

- (a) Determine the eigenvalues and normalized eigenvectors of the matrix A :

$$A = \begin{pmatrix} 1 & -i \\ i & 3 \end{pmatrix}$$

[8]

- (b) Suppose $R = (a_{ij})$ is an orthogonal matrix that rotates the vector $\vec{r}$ to $\vec{r}'$ (so that $x'_i = a_{ij}x_j$). From the invariance of the square of the length of the vector, derive the orthogonality condition:

$$\sum_{k=1}^3 a_{ik}a_{jk} = \delta_{ij} \quad \text{for } i, j = 1, 2, 3$$

[6]

- (c) How are the covariant and contravariant components of a vector related? A covariant tensor of rank 1 has components xy , $(y^2 - z^2)$, and x in rectangular Cartesian coordinates. Determine its components in spherical polar coordinates. [6]

Question 5.

- (a) Consider the periodic function $f(t)$ with period $T = 4$:

$$f(t) = \begin{cases} 0, & -2 < t < -1 \\ h, & -1 < t < 1 \\ 0, & 1 < t < 2 \end{cases}$$

with $h > 0$.

- (i) Make a neat sketch of this function. [2]
 (ii) Find the Fourier series expansion of $f(t)$. [5]
 (iii) Determine the values of $f(t)$ at $t = 0$ and $t = 1$ as given by the Fourier series. [3]
- (b) Legendre's differential equation $(1 - x^2)y'' - 2xy' + n(n + 1)y = 0$ has a regular solution $P_n(x)$ and a linearly independent irregular solution $Q_n(x)$. Show that their Wronskian is:

$$W(x) = P_n(x)Q'_n(x) - Q_n(x)P'_n(x) = \frac{A_n}{1 - x^2}$$

where A_n is a constant independent of x . [10]